

Jacob Helwig

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Education

Texas A&M University, Ph.D. Computer Science	August 2021-May 2026
The University of Texas at Austin, BS Mathematics	August 2016-May 2021

Work Experience

LinkedIn, Senior Software Engineer, Machine Learning	June 2026 - Present
• Reinforcement Learning for Agentic Search	
LinkedIn, CoreAI Intern	Summer 2024, Summer 2025
• Developed full training pipeline for large-scale sequential recommendation models.	
Los Alamos National Laboratory, Applied ML Fellow	Summer 2023
• Project: Solving the Elastic Wave Equation Using Deep Learning	

OSS Contributions

verl, Volcano Engine Reinforcement Learning for LLMs ([All Merged PRs](#))

- [On-Policy Distillation](#)
- [Multi-Teacher On-Policy Distillation](#)

Publications and Talks

[Preprint]	Data-Efficient Autoregressive-to-Diffusion Language Models via On-Policy Distillation
	Co-first author
[ICML 2026]	Learnability-Informed Fine-Tuning of Diffusion Language Models
	Paper
[ICLR 2026]	A Two-Phase Deep Learning Framework for Adaptive Time-Stepping in High-Speed Flow Modeling
	First author, paper , code
[NeurIPS 2024]	A Geometry-Aware Message Passing Neural Network for Modeling Aerodynamics over Airfoils (talk at the NeurIPS 2024 ML4CFD Competition)
	Best student solution, paper , code
[ICML 2024]	Equivariance via Minimal Frame Averaging for More Symmetries and Efficiency
	Spotlight, paper
[FnT ML]	Artificial Intelligence for Science in Quantum, Atomistic, and Continuum Systems
	Co-first author, paper
[ICLR 2024]	SineNet: Learning Temporal Dynamics in Time-Dependent PDEs
	Co-first author, paper , code
[LoG 2023]	High-fidelity Fluid Flow Reconstruction
	Paper
[ICML 2023]	Group Equivariant Fourier Neural Operators for Partial Differential Equations
	Co-first author, paper , code
[ACM TOMS]	A Covariate-Dependent Approach to Gaussian Graphical Modeling in R
	First author, paper , R package
[Preprint]	An Approximate Bayesian Approach to Covariate-dependent Graphical Modeling
	Paper

Honors

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- Ruth J. & Howard F. Newton Memorial Graduate Student Teaching Award In Statistics, 2022 Recipient
- “Jacob was selected to receive the Newton Teaching Award based on his outstanding evaluations by the instructors he was Teaching Assistant for during the Fall 2021 and Spring 2022 semesters.” ([award details](#))